Problem Set 72

Indeterminate Forms and L'Hopital's Rule

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Problem 4

Find the limit. Use L'Hopital's Rule where appropriate. If there is a more elementary method, consider using it. If L'Hoptial's Rule doesn't apply, explain why.

(c) $\lim_{x\to 0} \frac{\tan px}{\tan qx}$

$$\begin{cases} \lim_{x\to 0} \tan px = 0 \\ \lim_{x\to 0} \tan qx = 0 \end{cases}$$

$$\lim_{x\to 0} \frac{\tan px}{\tan qx} \overset{[\frac{0}{0}]}{=} \lim_{x\to 0} \frac{p \sec^2 px}{q \sec^2 qx} = \boxed{\frac{p}{q}}$$

(d) $\lim_{x\to\infty} \frac{\ln x}{\sqrt{x}}$

$$\begin{cases} \lim_{x\to\infty} \ln x \to \infty \\ \lim_{x\to\infty} \sqrt{x} \to \infty \end{cases}$$

$$\lim_{x\to\infty} \frac{\ln x}{\sqrt{x}} \overset{[\frac{\infty}{\infty}]}{=} \lim_{x\to\infty} \frac{\frac{1}{x}}{\frac{1}{2\sqrt{x}}} = \lim_{x\to\infty} \frac{2}{\sqrt{x}} = \boxed{0}$$

(e) $\lim_{x\to 0^+} \frac{\ln x}{x}$

$$\begin{cases} \lim_{x\to 0^+} \ln x \to -\infty \\ \lim_{x\to 0^+} \frac{1}{x} \to \infty \end{cases}$$

$$\lim_{x\to 0^+} \frac{\ln x}{x} \to \boxed{-\infty}$$

(f) $\lim_{x\to 0} \frac{e^x}{x^2}$

$$\begin{cases} \lim_{x\to 0^+} e^x = 1 \\ \lim_{x\to 0^+} x^2 = 0 \end{cases}$$

$$\lim_{x\to 0} \frac{e^x}{x^2} = \boxed{0}$$

(g) $\lim\limits_{t\to 0} \dfrac{5^t - 3^t}{t}$

$$\begin{cases} \lim\limits_{t\to 0} 5^t - 3^t = 0 \\ \lim\limits_{t\to 0} t = 0 \end{cases}$$

$$\lim_{t\to 0} \frac{5^t - 3^t}{t} \overset{[\frac{0}{0}]}{=} \lim_{t\to 0} \frac{5^t \ln 5 - 3^t \ln 3}{1} = \boxed{\ln 5 - \ln 3}$$

(h) $\lim\limits_{x\to 0} \dfrac{1 - \cos x}{x^2}$

$$\begin{cases} \lim\limits_{x\to 0} 1 - \cos x = 0 \\ \lim\limits_{x\to 0} x^2 = 0 \end{cases}$$

$$\lim_{x\to 0} \frac{1 - \cos x}{x^2} \overset{[\frac{0}{0}]}{=} \lim_{x\to 0} \frac{\sin x}{2x} = \frac{1}{2} \lim_{x\to 0} \frac{\sin x}{x} = \boxed{\frac{1}{2}}$$

(i) $\lim\limits_{x\to 0} \dfrac{x + \sin x}{x + \cos x}$

$$\lim_{x\to 0} \frac{x + \sin x}{x + \cos x} = \lim_{x\to 0} \frac{0}{1} = \boxed{0}$$

(j) $\lim\limits_{x\to 1} \dfrac{1 - x + \ln x}{1 + \cos \pi x}$

$$\begin{cases} \lim\limits_{x\to 1} 1 - x + \ln x = 0 \\ \lim\limits_{x\to 1} 1 + \cos \pi x = 0 \end{cases}$$

$$\lim_{x\to 1} \frac{1 - x + \ln x}{1 + \cos \pi x} \overset{[\frac{0}{0}]}{=} \lim_{x\to 1} \frac{\frac{1}{x} - 1}{-\pi \sin \pi x}$$

$$\begin{cases} \lim\limits_{x\to 1} \dfrac{1}{x} - 1 = 0 \\ \lim\limits_{x\to 1} -\pi \sin \pi x = 0 \end{cases}$$

$$\lim_{x\to 1} \frac{\frac{1}{x} - 1}{-\pi \sin \pi x} = \lim_{x\to 1} \frac{-\frac{1}{x^2}}{-\pi^2 \cos \pi} = \boxed{-\frac{1}{\pi^2}}$$

(k) $\lim\limits_{x\to 1} \dfrac{x^a - ax + a - 1}{(x-1)^2}$

$$\begin{cases} \lim\limits_{x\to 1} x^a - ax + a - 1 = 0 \\ \lim\limits_{x\to 1} (x-1)^2 = 0 \end{cases}$$

$$\lim_{x\to 1} \frac{x^a - ax + a - 1}{(x-1)^2} \overset{[\frac{0}{0}]}{=} \lim_{x\to 1} \frac{ax^{a-1} - a}{2(x-1)}$$

$$\begin{cases} \lim_{x\to 1} ax^{a-1} - a = 0 \\ \lim_{x\to 1} 2(x-1) = 0 \end{cases}$$

$$\lim_{x\to 1} \frac{ax^{a-1} - a}{2x-2} \overset{[\frac{0}{0}]}{=} \lim_{x\to 1} \frac{a(a-1)x^{a-2}}{2} = \boxed{\frac{a^2-a}{2}}$$

(l) $\lim_{x\to\infty} x \sin \frac{\pi}{x}$

$$\lim_{x\to\infty} x \sin \frac{\pi}{x} = \lim_{x\to\infty} \frac{\sin \frac{\pi}{x}}{\frac{\pi}{x}} \cdot \pi = \boxed{\pi}$$

(m) $\lim_{x\to 0} \cot 2x \sin 6x$

$$\lim_{x\to 0} \frac{\cos 2x \sin 6x}{\sin 2x}$$

$$\begin{cases} \lim_{x\to 0} \cos 2x \sin 6x = 0 \\ \lim_{x\to 0} \sin 2x = 0 \end{cases}$$

$$\lim_{x\to 0} \frac{\cos 2x \sin 6x}{\sin 2x} \overset{[\frac{0}{0}]}{=} \lim_{x\to 0} \frac{6\cos 2x \cos 6x - 2\sin 2x \sin 6x}{\cos 2x} = \boxed{6}$$

(n) $\lim_{x\to\infty} x^3 e^{-x^2}$

$$\lim_{x\to\infty} \frac{x^3}{e^{x^2}}$$

$$\begin{cases} \lim_{x\to\infty} x^3 \to \infty \\ \lim_{x\to\infty} e^{x^2} \to \infty \end{cases}$$

$$\lim_{x\to\infty} \frac{x^3}{e^{x^2}} \overset{[\frac{\infty}{\infty}]}{=} \lim_{x\to\infty} \frac{3x^2}{2xe^{x^2}} = \lim_{x\to\infty} \frac{3x}{2e^{x^2}}$$

$$\begin{cases} \lim_{x\to\infty} 3x \to \infty \\ \lim_{x\to\infty} 2e^{x^2} \to \infty \end{cases}$$

$$\lim_{x\to\infty} \frac{3x}{2e^{x^2}} \overset{[\frac{\infty}{\infty}]}{=} \lim_{x\to\infty} \frac{3}{4xe^{x^2}} = \boxed{0}$$

(o) $\lim_{x\to 1^+} \ln x \tan \frac{\pi x}{2}$

$$\lim_{x\to 1^+} \ln x \tan \frac{\pi x}{2} = \lim_{x\to 1^+} \frac{\ln x \sin \frac{\pi x}{2}}{\cos \frac{\pi x}{2}}$$

$$\begin{cases} \lim_{x\to 1^+} \ln x \sin \frac{\pi x}{2} = 0 \\ \lim_{x\to 1^+} \cos \frac{\pi x}{2} = 0 \end{cases}$$

$$\lim_{x\to 1^+} \frac{\ln x \sin \frac{\pi x}{2}}{\cos \frac{\pi x}{2}} \overset{[\frac{0}{0}]}{=} \lim_{x\to 1^+} \frac{\frac{\sin \frac{\pi x}{2}}{x} + \frac{\pi}{2} \ln x \cos \frac{\pi x}{2}}{-\frac{\pi}{2} \sin \frac{\pi x}{2}} = \boxed{-\frac{2}{\pi}}$$

(p) $\lim_{x\to 1}\left(\frac{x}{x-1}-\frac{1}{\ln x}\right)$

$$\lim_{x\to 1}\left(\frac{x}{x-1}-\frac{1}{\ln x}\right)=\lim_{x\to 1}\frac{x\ln x-x+1}{(x-1)\ln x}$$

$$\begin{cases}\lim\limits_{x\to 1}x\ln x-x+1=0\\ \lim\limits_{x\to 1}(x-1)\ln x=0\end{cases}$$

$$\lim_{x\to 1}\frac{x\ln x-x+1}{(x-1)\ln x}\overset{[\frac{0}{0}]}{=}\lim_{x\to 1}\frac{\ln x}{\ln x+\frac{(x-1)}{x}}$$

$$\begin{cases}\lim\limits_{x\to 1}\ln x=0\\ \lim\limits_{x\to 1}\ln x+\frac{(x-1)}{x}=0\end{cases}$$

$$\lim_{x\to 1}\frac{\ln x}{\ln x+\frac{(x-1)}{x}}\overset{[\frac{0}{0}]}{=}\lim_{x\to 1}\frac{\frac{1}{x}}{\frac{1}{x}+\frac{1}{x^2}}=\boxed{\frac{1}{2}}$$

(q) $\lim_{x\to\infty}\left(\sqrt{x^2+x}-x\right)$

$$\lim_{x\to\infty}\left(\sqrt{x^2+x}-x\right)=\lim_{x\to\infty}\frac{x}{\sqrt{x^2+x}+x}=\lim_{x\to\infty}\frac{1}{\sqrt{1+\frac{1}{x}}+1}=\boxed{\frac{1}{2}}$$